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Лекция Использование Terraform в команде

Задание 0

1. Прочтите статью: <https://neprivet.com/>
2. Пожалуйста, распространите данную идею в своем коллективе.

Исполнено. Максимально поддерживаю и придерживаюсь аналогичных взглядов на стили общения.

Задание 1

1. Возьмите код:
 - из [ДЗ к лекции 4](#),
 - из [демо к лекции 4](#).
2. Проверьте код с помощью tflint и checkov. Вам не нужно инициализировать этот проект.
3. Перечислите, какие **типы** ошибок обнаружены в проекте (без дублей).

Разбор решения задачи 1

Устанавливаю инструменты:

```
1  curl -s https://raw.githubusercontent.com/terraform-linters/tflint/master/install_linux.sh | bash tflint --  
    version  
2  
3  # установка в виртуальном окружении с добавлением .venv в gitignore  
4  
5  sudo apt install python3-pip  
6  python3 -m venv .venv  
7  source .venv/bin/activate  
8  pip install checkov  
9
```

Запустил:

```
1  checkov -d /home/vboxuser/terraform/hw-04
```

```

103 2.7.7.0 width 0.7.0 xmethod 1.0.1 junc 1.2.1.2 zipp 4.1.0
• (.venv) vboxuser@KVA:~/terraform/hw-05$ checkov --version
3.2.529
⊙ (.venv) vboxuser@KVA:~/terraform/hw-05$ checkov -d /home/vboxuser/terraform/hw-04
2026-05-26 13:33:56,691 [MainThread ] [WARNI] Failed to get the checkov mappings and guidelines from https://api0.prismacloud.io/bridgecrew/api/
v2/guidelines. Skips using BC * IDs will not work.
Traceback (most recent call last):
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connection.py", line 204, in _new_conn
    sock = connection.create_connection(
            ~~~~~
            ~~~~~
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/util/connection.py", line 85, in create_connection
    raise err
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/util/connection.py", line 73, in create_connection
    sock.connect(sa)
TimeoutError: timed out

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 788, in urlopen
    response = self.make_request(
            ~~~~~
            ~~~~~
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 488, in _make_request
    raise new_e
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 464, in _make_request
    self.validate_conn(conn)
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 1106, in _validate_conn
    conn.connect()
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connection.py", line 759, in connect
    self.sock = sock = self._new_conn()
            ~~~~~
            ~~~~~
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/connection.py", line 213, in _new_conn
    raise ConnectTimeoutError(
urllib3.exceptions.ConnectTimeoutError: (<HTTPSConnection(host='api0.prismacloud.io', port=443) at 0x764fa1b9e990>, 'Connection to api0.prismacloud.io timed out. (connect timeout=3.1)')

The above exception was the direct cause of the following exception:

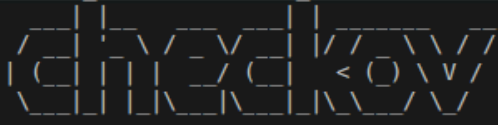
Traceback (most recent call last):
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/checkov/common/bridgecrew/platform_integration.py", line 1276, in get_public_run_config
    request = self.http.request("GET", self.guidelines_api_url, headers=headers) # type:ignore[no-untyped-call]
            ~~~~~
            ~~~~~
  File "/home/vboxuser/terraform/hw-05/.venv/lib/python3.12/site-packages/urllib3/request methods.py", line 135, in request

```

```

File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/_request_methods.py", line 135, in request
    return self.request_encode_url(
           ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/_request_methods.py", line 182, in request_encode_url
    return self.urlopen(method, url, **extra_kw)
           ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/poolmanager.py", line 457, in urlopen
    response = conn.urlopen(method, u.request_uri, **kw)
               ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/connectionpool.py", line 872, in urlopen
    return self.urlopen(
           ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/connectionpool.py", line 872, in urlopen
    return self.urlopen(
           ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/connectionpool.py", line 872, in urlopen
    return self.urlopen(
           ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/connectionpool.py", line 842, in urlopen
    retries = retries.increment(
           ~~~~~~
File "/home/vboxuser/terraform/hw-05/.env/lib/python3.12/site-packages/urllib3/util/retry.py", line 543, in increment
    raise MaxRetryError(_pool, url, reason) from reason # type: ignore[arg-type]
           ~~~~~~
urllib3.exceptions.MaxRetryError: HTTPSConnectionPool(host='api0.prismacloud.io', port=443): Max retries exceeded with url: /bridgecrew/api/v2/guidelines (Caused by ConnectTimeoutError(<HTTPSConnection(host='api0.prismacloud.io', port=443) at 0x764fa1b9e990>, 'Connection to api0.prismacloud.io timed out. (connect timeout=3.1)'))
[kubernetes framework ]: 100%|██████████| [2/2], Current File Scanned=vault/terraform/docker-compose.yml
[secrets framework ]: 0%|██████████| [0/18]2026-05-26 13:33:57,039 [MainThread ] [WARNI] Failed to download module git::https://github.com/udjin10/yandex compute instance.git?ref=main (for external modules, the --download-external-modules flag is required)
2026-05-26 13:33:57,054 [MainThread ] [WARNI] Unable to load module (github.com/terraform-yc-modules/terraform-yc-s3): list index out of range
[secrets framework ]: 100%|██████████| [18/18], Current File Scanned=/home/vboxuser/terraform/hw-04/s3/versions.tf
[ansible framework ]: 100%|██████████| [1/1], Current File Scanned=cloud-init.yml/main.tf
[terraform framework ]: 100%|██████████| [15/15], Current File Scanned=../hw-04/vpc/variables.tf

```



By Prisma Cloud | version: 3.2.529

terraform scan results:

Passed checks: 1, Failed checks: 6, Skipped checks: 0

Check: CKV_AWS_41: "Ensure no hard coded AWS access key and secret key exists in provider"

PASSED for resource: aws.default

File: /s3/providers.tf:8-15

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: marketing_vm

File: /main.tf:32-52

```
32 | module "marketing_vm" {
33 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
34 |   env_name    = "develop"
35 |   network_id  = module.vpc_dev.network_id
36 |   subnet_zones = [var.default_zone]
37 |   subnet_ids  = [module.vpc_dev.subnets["ru-central1-a"].id]
38 |   instance_name = "marketing"
39 |   instance_count = 1
40 |   image_family  = "ubuntu-2004-lts"
41 |   public_ip     = false
42 |
43 |   labels = {
44 |     owner  = "team-marketing"
45 |     project = "marketing"
46 |   }
47 |
48 |   metadata = {
49 |     user-data      = data.template_file.cloudinit.rendered
50 |     serial-port-enable = 1
51 |   }
52 | }
```

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: marketing_vm

File: /main.tf:32-52

```

32 | module "marketing_vm" {
33 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
34 |   env_name    = "develop"
35 |   network_id  = module.vpc_dev.network_id
36 |   subnet_zones = [var.default_zone]
37 |   subnet_ids  = [module.vpc_dev.subnets["ru-central1-a"].id]
38 |   instance_name = "marketing"
39 |   instance_count = 1
40 |   image_family = "ubuntu-2004-lts"
41 |   public_ip    = false
42 |
43 |   labels = {
44 |     owner  = "team-marketing"
45 |     project = "marketing"
46 |   }
47 |
48 |   metadata = {
49 |     user-data      = data.template_file.cloudinit.rendered
50 |     serial-port-enable = 1
51 |   }
52 | }

```

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: analytics_vm

File: /main.tf:54-74

```

54 | module "analytics_vm" {
55 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
56 |   env_name    = "develop"
57 |   network_id  = module.vpc_dev.network_id
58 |   subnet_zones = [var.default_zone]
59 |   subnet_ids  = [module.vpc_dev.subnets["ru-central1-a"].id]
60 |   instance_name = "analytics"
61 |   instance_count = 1
62 |   image_family = "ubuntu-2004-lts"
63 |   public_ip    = false
64 |
65 |   labels = {
66 |     owner  = "team-analytics"
67 |     project = "analytics"
68 |   }

```

```

69 |
70 |     metadata = {
71 |       user-data      = data.template_file.cloudinit.rendered
72 |       serial-port-enable = 1
73 |     }
74 |   }

```

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: analytics_vm
File: /main.tf:54-74

```

54 | module "analytics_vm" {
55 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
56 |   env_name    = "develop"
57 |   network_id  = module.vpc_dev.network_id
58 |   subnet_zones = [var.default_zone]
59 |   subnet_ids  = [module.vpc_dev.subnets["ru-central1-a"].id]
60 |   instance_name = "analytics"
61 |   instance_count = 1
62 |   image_family  = "ubuntu-2004-lts"
63 |   public_ip     = false
64 |
65 |   labels = {
66 |     owner  = "team-analytics"
67 |     project = "analytics"
68 |   }
69 |
70 |   metadata = {
71 |     user-data      = data.template_file.cloudinit.rendered
72 |     serial-port-enable = 1
73 |   }
74 | }

```

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: s3
File: /s3/main.tf:1-7

```

1 | module "s3" {
2 |   source = "github.com/terraform-yc-modules/terraform-yc-s3"
3 |
4 |   bucket_name = "netology-course-bucket-kva" # уникальное имя по всему облаку
5 |   force_destroy = false                     # флаг не удалять с объектами
6 |   max_size      = 1073741824                 # размер 1 ГБ в байтах
7 | }

```

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: s3

```

1 | module "s3" {
2 |   source = "github.com/terraform-yc-modules/terraform-yc-s3"
3 |
4 |   bucket_name = "netology-course-bucket-kva" # уникальное имя по всему облаку
5 |   force_destroy = false                      # флаг не удалять с объектами
6 |   max_size     = 1073741824                  # размер 1 ГБ в байтах
7 | }

```

secrets scan results:

Passed checks: 0, Failed checks: 2, Skipped checks: 0

Check: CKV_SECRET_6: "Base64 High Entropy String"
 FAILED for resource: 24e7451df05ed5cd4cf1041be67c68f8d89d087a
 File: /vault/terraform/main.tf:4-5

```

4 |   token = "ed*****"

```

Check: CKV_SECRET_6: "Base64 High Entropy String"
 FAILED for resource: 4701dd25f0adb67ab0c84e46633be8b4e7e6e44f
 File: /vault/terraform/main.tf:15-16

```

15 |   password = "supe*****"

```

• (.venv) vboxuser@KVA:~/terraform/hw-05\$ deactivate

Checkov нашёл **3** типа проблем:

1. CKV_TF_1: Модуль подключён не по commit hash

1 Ensure Terraform module sources use a commit hash

Найдено в: `marketing_vm`, `analytics_vm`, `s3`

Суть : Все три модуля подключены по имени ветки (`?ref=main`) или без версии вообще. Это небезопасно, так как со сменой кода в `main` `terraform init` может притащить зловреда.

Правильно указывать конкретный commit hash:

`"git::https://github.com/udjin10/yandex_compute_instance.git?ref=a1b2c3d4e5f6..."`

2. CKV_TF_2: Модуль подключён не по versionному тегу

```
1 Ensure Terraform module sources use a tag with a version number
```

Найдено в: `marketing_vm`, `analytics_vm`, `s3`

Суть: Нет тега версии типа `v1.0.0`. Это более лайтовая версия предыдущей проблемы, связанной с использованием хотябы тегированной версии, если нет commit hash:

```
1 git::[https://github.com/udjin10/yandex_compute_instance.git?ref=v1.2.0]
  (https://github.com/udjin10/yandex_compute_instance.git?ref=v1.2.0)
```

3. CKV_SECRET_6: Секреты захардкожены в коде

```
1 Base64 High Entropy String
```

Найдено в: `vault/terraform/main.tf`, токен Vault (`education`) и пароль (`superpassword123`) прямо в `.tf`-файле.

Суть: Секреты не должны храниться в коде в открытом виде. Правильно передавать через переменные окружения или `terraform.tfvars`

Запустил tfllint из папки с hw-04 - с кодом предыдущего ДЗ:

```
1 tfllint --recursive
```

```
⊙ vboxuser@KVA:~/terraform/hw-04$ tflint --recursive
11 issue(s) found:

Warning: Module source "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main" uses a default branch as ref (main) (terraform_module_pinned_source)

  on main.tf line 33:
  33:   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: Module source "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main" uses a default branch as ref (main) (terraform_module_pinned_source)

  on main.tf line 55:
  55:   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: Missing version constraint for provider "yandex" in `required_providers` (terraform_required_providers)

  on providers.tf line 3:
   3:     yandex = {
   4:       source = "yandex-cloud/yandex"
   5:     }

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

Warning: Module source "github.com/terraform-yc-modules/terraform-yc-s3" is not pinned (terraform_module_pinned_source)

  on s3/main.tf line 2:
   2:   source = "github.com/terraform-yc-modules/terraform-yc-s3"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: [Fixable] variable "default_cidr" is declared but not used (terraform_unused_declarations)

  on variables.tf line 24:
```

```
24: variable "default_cidr" {
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: [Fixable] variable "vpc_name" is declared but not used (terraform_unused_declarations)

on variables.tf line 30:

```
30: variable "vpc_name" {
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: [Fixable] variable "vm_web_name" is declared but not used (terraform_unused_declarations)

on variables.tf line 46:

```
46: variable "vm_web_name" {
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: [Fixable] variable "vm_db_name" is declared but not used (terraform_unused_declarations)

on variables.tf line 53:

```
53: variable "vm_db_name" {
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: terraform "required_version" attribute is required (terraform_required_version)

on vault/terraform/versions.tf line 1:

```
1: terraform {
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_version.md

Warning: terraform "required_version" attribute is required (terraform_required_version)

on vpc/providers.tf line 1:

```
1: terra
```

[Follow link \(ctrl + click\)](#)

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_version.md

Warning: Missing version constraint for provider "yandex" in `required\_providers` (terraform_required_providers)

on vpc/providers.tf line 3:

```
3:   yandex = {
```

```
4:     source = "yandex-cloud/yandex"
```

```
5: ____}
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

```
o vboxuser@KVA:~/terraform/hw-04$ ^C
o vboxuser@KVA:~/terraform/hw-04$ █
```

tflint нашёл 4 типа предупреждений:

1. `terraform_module_pinned_source`
Модуль подключён к ветке `main` без версии. Это небезопасно, по той же причине что и описано выше.
Найдено в: `main.tf` (`marketing_vm`, `analytics_vm`), `s3/main.tf`.
2. `terraform_required_providers` - не указана версия провайдера `yandex` в блоке `required_providers`.
Найдено в: `providers.tf`, `vpc/providers.tf`
3. `terraform_unused_declarations` - объявлены переменные, которые нигде не используются: `default_cidr`, `vpc_name`, `vm_web_name`, `vm_db_name`
Найдено в: `variables.tf`
4. `terraform_required_version` - отсутствует атрибут `required_version` в блоке `terraform {}` — не зафиксирована минимальная версия Terraform.

Проверка `tflint` кода из git clone <https://github.com/netology-code/ter-homeworks/tree/main/04/demonstration1>:

```
vboxuser@KVA:~/terraform/ter-homeworks/04/demonstration1$ tflint --recursive
6 issue(s) found:

Warning: Missing version constraint for provider "random" in `required_providers` (terraform_required_providers)

  on passwords/main.tf line 8:
   8: resource "random_password" "input_vms" {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

Warning: Module source "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main" uses a default branch as ref (main) (terraform_module_pinned_source)

  on vms/main.tf line 23:
  23: source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: Module source "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main" uses a default branch as ref (main) (terraform_module_pinned_source)

  on vms/main.tf line 46:
  46: source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: Missing version constraint for provider "template" in `required_providers` (terraform_required_providers)

  on vms/main.tf line 64:
  64: data "template_file" "cloudinit" {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

Warning: Missing version constraint for provider "yandex" in `required_providers` (terraform_required_providers)

  on vms/providers.tf line 3:
   3:   yandex = {
   4:     source = "yandex-cloud/yandex"
   5:   }

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md
```

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

Warning: [Fixable] variable "public_key" is declared but not used (terraform_unused_declarations)

on vms/variables.tf line 3:
3: `variable "public_key" {`

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

`vboxuser@KVA:~/terraform/ter-homeworks/04/demonstration1$`

`tflint` в учебном коде (demonstration1) нашел 6 предупреждений/ошибок:

Тип	Правило	Где
Модуль без фиксированной версии (<code>?ref=main</code>)	<code>terraform_module_pinned_source</code>	<code>vms/main.tf</code> (2 раза)
Не указана версия провайдера	<code>terraform_required_providers</code>	<code>random</code> , <code>template</code> , <code>yandex</code>
Неиспользуемая переменная	<code>terraform_unused_declarations</code>	<code>vms/variables.tf</code> - <code>public_key</code>

Запустил `Chekov`

```

(.venv) vboxuser@KVA:~/terraform/ter-homeworks/04/demonstration1$ checkov -d ~/terraform/ter-homeworks/04/demonstration1
2026-05-26 17:18:28,996 [MainThread ] [WARNI] Failed to get the checkov mappings and guidelines from https://api0.prismacloud.io/bridgecrew/api/v2/guidelines. Skips using BC * IDs will not work.
Traceback (most recent call last):
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connection.py", line 204, in _new_conn
    sock = connection.create_connection(
           ~~~~~
           ~~~~~
           ~~~~~
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/util/connection.py", line 85, in create_connection
    raise err
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/util/connection.py", line 73, in create_connection
    sock.connect(sa)
TimeoutError: timed out

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 788, in urlopen
    response = self._make_request(
               ~~~~~
               ~~~~~
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 488, in _make_request
    raise new_e
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 464, in _make_request
    self._validate_conn(conn)
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 1106, in _validate_conn
    conn.connect()
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connection.py", line 759, in connect
    self.sock = sock = self._new_conn()
                  ~~~~~
  File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connection.py", line 213, in _new_conn
    raise ConnectTimeoutError(
urllib3.exceptions.ConnectTimeoutError: (<HTTPSConnection(host='api0.prismacloud.io', port=443) at 0x70da0064bb30>, 'Connection to api0.prismacloud.io timed out. (connect timeout=3.1)')

```

The above exception was the direct cause of the following exception:

Traceback (most recent call last):

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/checkov/common/bridgecrew/platform_integration.py", line 1276, in get_public_run_config
```

```
request = self.http.request("GET", self.guidelines_api_url, headers=headers) # type: ignore[no-untyped-call]
```

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/_request_methods.py", line 135, in request
```

```
return self.request_encode_url(
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/_request_methods.py", line 182, in request
    encode_url
```

```
return self.urlopen(method, url, **extra_kw)
#####
```

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/poolmanager.py", line 457, in urlopen
    response = conn.urlopen(method, u.request_uri, **kw)
```

~~~~~

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 872, in urlopen
    en
```

```
return self.urlopen(
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 872, in urlopen
    en
```

```
return self.urlopen(
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 872, in urlopen
    en
```

```
return self.urlopen(
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/connectionpool.py", line 842, in urlopen
    en
```

```
retries = retries.increment(
    ^^^^^^^^^^^^^^^^^^^^^^^^^
```

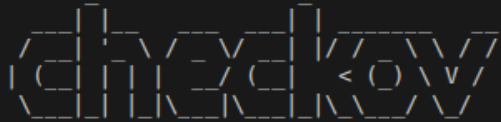
```
File "/home/vboxuser/terraform/ter-homeworks/04/demonstration1/.venv/lib/python3.12/site-packages/urllib3/util/retry.py", line 543, in increment
    raise MaxRetryError(pool, url, reason) from reason # type: ignore[arg-type]
```

~~~~~

```
urllib3.exceptions.MaxRetryError: HTTPConnectionPool(host='api0.prismacloud.io', port=443): Max retries exceeded with url: /bridgecrew/api/v2/guidelines (Caused by ConnectTimeoutError(<HTTPConnection(host='api0.prismacloud.io', port=443) at 0x70da0064bb30>, 'Connection to api0.prismacloud.io timed out. (connect timeout=3.1)'))
```

```
[ kubernetes framework ]: 100%|██████████| [1/1], Current File Scanned=vms/cloud-init.yml
```

```
2026-05-26 17:18:29,080 [MainThread ] [WARNI] Failed to download module git::https://github.com/udjin10/yandex_compute_instance.git?ref=main (for external modules, the --download-external-modules flag is required)
[ terraform framework ]: 100%|██████████| [7/7], Current File Scanned=vms/variables.tf
[ ansible framework ]: 100%|██████████| [1/1], Current File Scanned=vms/cloud-init.ymlraform/ter-homeworks/04/demonstration1/vms/remote_s
[ secrets framework ]: 100%|██████████| [8/8], Current File Scanned=/home/vboxuser/terraform/ter-homeworks/04/demonstration1/vms/provider
```



By Prisma Cloud | version: 3.2.529

terraform scan results:

Passed checks: 0, Failed checks: 4, Skipped checks: 0

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: test-vm

File: /vms/main.tf:22-43

```
22 | module "test-vm" {
23 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
24 |   env_name    = "develop"
25 |   network_id  = yandex_vpc_network.develop.id
26 |   subnet_zones = ["ru-central1-a", "ru-central1-b"]
27 |   subnet_ids  = [yandex_vpc_subnet.develop_a.id, yandex_vpc_subnet.develop_b.id]
28 |   instance_name = "webs"
29 |   instance_count = 2
30 |   image_family  = "ubuntu-2004-lts"
31 |   public_ip     = true
32 |
33 |   labels = {
34 |     owner= "i.ivanov",
35 |     project = "accounting"
36 |   }
37 |
38 |   metadata = {
39 |     user-data      = data.template_file.cloudinit.rendered #Для демонстрации №3
40 |     serial-port-enable = 1
41 |   }
```

43 | }

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: test-vm

File: /vms/main.tf:22-43

```
22 | module "test-vm" {
23 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
24 |   env_name    = "develop"
25 |   network_id  = yandex_vpc_network.develop.id
26 |   subnet_zones = ["ru-central1-a", "ru-central1-b"]
27 |   subnet_ids  = [yandex_vpc_subnet.develop_a.id, yandex_vpc_subnet.develop_b.id]
28 |   instance_name = "webs"
29 |   instance_count = 2
30 |   image_family = "ubuntu-2004-lts"
31 |   public_ip    = true
32 |
33 |   labels = {
34 |     owner = "i.ivanov",
35 |     project = "accounting"
36 |   }
37 |
38 |   metadata = {
39 |     user-data      = data.template_file.cloudinit.rendered #Для демонстрации №3
40 |     serial-port-enable = 1
41 |   }
42 |
43 | }
```

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: example-vm

File: /vms/main.tf:45-61

```
45 | module "example-vm" {
46 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
47 |   env_name    = "stage"
48 |   network_id  = yandex_vpc_network.develop.id
49 |   subnet_zones = ["ru-central1-a"]
50 |   subnet_ids  = [yandex_vpc_subnet.develop_a.id]
51 |   instance_name = "web-stage"
52 |   instance_count = 1
53 |   image_family = "ubuntu-2004-lts"
54 |   public_ip    = true
```

```

55 |
56 |   metadata = {
57 |     user-data      = data.template_file.cloudinit.rendered #Для демонстрации №3
58 |     serial-port-enable = 1
59 |   }
60 |
61 | }

```

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: example-vm

File: /vms/main.tf:45-61

```

45 | module "example-vm" {
46 |   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"
47 |   env_name    = "stage"
48 |   network_id  = yandex_vpc_network.develop.id
49 |   subnet_zones = ["ru-central1-a"]
50 |   subnet_ids  = [yandex_vpc_subnet.develop_a.id]
51 |   instance_name = "web-stage"
52 |   instance_count = 1
53 |   image_family = "ubuntu-2004-lts"
54 |   public_ip    = true
55 |
56 |   metadata = {
57 |     user-data      = data.template_file.cloudinit.rendered #Для демонстрации №3
58 |     serial-port-enable = 1
59 |   }
60 |
61 | }

```

(.venv) vboxuser@KVA:~/terraform/ter-homeworks/04/demonstration1\$

Итого, общее количество уникальных ошибок в обеих папках проектов:

	Тип предупреждения/ошибки	Правило	Инструмент	Найдено в
1	Модуль подключён по ветке <code>main</code> вместо commit hash	<code>CKV_TF_1</code> , <code>terraform_module_pinned_source</code>	checkov + tflint	hw-04, demonstration1
2	Модуль подключён без версионного тега	<code>CKV_TF_2</code>	checkov	hw-04, demonstration1
3	Не указана версия провайдера в <code>required_providers</code>	<code>terraform_required_providers</code>	tflint	hw-04, demonstration1

	Тип предупреждения/ошибки	Правило	Инструмент	Найдено в
4	Отсутствует <code>required_version</code> в блоке <code>terraform {}</code>	<code>terraform_required_version</code>	tflint	hw-04
5	Объявлены, но не используются переменные	<code>terraform_unused_declarations</code>	tflint	hw-04, demonstration1
6	Секреты захардкожены прямо в коде	<code>CKV_SECRET_6</code>	checkov	hw-04 (vault)

Ссылка на документацию: <https://github.com/terraform-linters/tflint?tab=readme-ov-file>
<https://www.checkov.io/>

Задание 2

1. Возьмите ваш GitHub-репозиторий с **выполненным ДЗ 4** в ветке 'terraform-04' и сделайте из него ветку 'terraform-05'.
2. Настройте remote state с встроенными блокировками:
 - Создайте S3 bucket в Yandex Cloud для хранения state (если еще не создан)
 - Создайте service account с правами на чтение/запись в bucket
 - Настройте backend в providers.tf с использованием нового механизма блокировок:

```

1  terraform {
2      required_version = "~>1.12.0"
3
4      backend "s3" {
5          bucket = "ваш-bucket-name"
6          key    = "terraform.tfstate"
7          region = "ru-central1"
8
9          # Встроенный механизм блокировок (Terraform >= 1.6)
10         # Не требует отдельной базы данных!

```

```

11     use_lockfile = true
12
13     endpoints = {
14         s3 = "https://storage.yandexcloud.net"
15     }
16
17     skip_region_validation      = true
18     skip_credentials_validation = true
19     skip_requesting_account_id  = true
20     skip_s3_checksum            = true
21 }
22 }

```

- Выполните `terraform init -migrate-state` для миграции state в S3
 - Предоставьте скриншоты процесса настройки и миграции
3. Закоммитьте в ветку 'terraform-05' все изменения.
 4. Откройте в проекте terraform console, а в другом окне из этой же директории попробуйте запустить terraform apply.
 5. Пришлите ответ об ошибке доступа к state (блокировка должна сработать автоматически).
 6. Принудительно разблокируйте state командой `terraform force-unlock <LOCK_ID>`. Пришлите команду и вывод.

Примечание: В Terraform >= 1.6 появился встроенный механизм блокировок через `use_lockfile = true`. Это упрощает настройку - больше не нужно создавать отдельную базу данных (YDB в режиме DynamoDB) для хранения блокировок. Lock-файл создается автоматически в том же S3 bucket рядом с state-файлом с именем `<key>.lock.info`.

Разбор решения задачи 2

Создал `service account` через `yc CLI`:

```

1 yc iam service-account create --name terraform-backend-sa
2 yc iam service-account list

```

```
• vboxuser@KVA:~/terraform/hw-05$ yc iam service-account create --name terraform-backend-sa
done (2s)
id: ajeg4nq635d5tm88f1ka
folder_id: blgrsjmidldjs2rmkgd0
created_at: "2026-05-26T17:33:49Z"
name: terraform-backend-sa
```

There is a new yc version '1.10.0' available. Current version: '1.1.0'.
See release notes at <https://yandex.cloud/ru/docs/cli/release-notes>
You can install it by running the following command in your shell:
\$ yc components update

```
• vboxuser@KVA:~/terraform/hw-05$ yc iam service-account list
```

ID	NAME	LABELS	CREATED AT	LAST AUTHENTICATED AT
ajec3mm6jkn7uleaum4p	scanner		2026-04-25 14:56:09	2026-04-25 15:30:00
ajeg4nq635d5tm88f1ka	terraform-backend-sa		2026-05-26 17:33:49	
ajer903546r5j0efpfls	kva-terraform		2026-05-18 11:24:11	2026-05-25 17:40:00

```
• vboxuser@KVA:~/terraform/hw-05$
```

Назначил права на bucket (роль `storage.editor`)

```
1 yc resource-manager folder add-access-binding \  
2   --id blgrsjmidldjs2rmkgd0 \  
3   --role storage.editor \  
4   --service-account-id ajeg4nq635d5tm88f1ka
```

```

• vboxuser@KVA:~/terraform/hw-05$ yc resource-manager folder add-access-binding \
  --id blgrsjmidldjs2rmkgd0 \
  --role storage.editor \
  --service-account-id ajeg4nq635d5tm88f1ka
done (2s)
effective_deltas:
- action: ADD
  access_binding:
    role_id: storage.editor
    subject:
      id: ajeg4nq635d5tm88f1ka
      type: serviceAccount

```

cloud-kost1n-valery  default    Identity and Access Management / Сервисные аккаунты

Сервисные аккаунты

Фильтр по имени или идентификатору

Имя	Роли в каталоге	Описание	Метки	Дата создания	Дата последней аутентификации	
scanner	container-registry.images.scanner	—	—	25.04.2026, в 17:56	25.04.2026	...
terraform-backend-sa	storage.editor	—	—	26.05.2026, в 20:33	26.05.2026	...
kva-terraform	editor	—	—	18.05.2026, в 14:24	25.05.2026	...

Создал статические ключи доступа:

```

1 yc iam access-key create \
2   --service-account-id ajeg4nq635d5tm88f1ka \
3   --description "terraform backend key"

```

```
• vboxuser@KVA:~/terraform/hw-05$ yc iam access-key create \
  --service-account-id ajeg4nq635d5tm88f1ka \
  --description "terraform backend key"
access_key:
  id: ajehrh9v6tk314ur8q6i
  service_account_id: ajeg4nq635d5tm88f1ka
  created_at: "2026-05-26T17:40:37.667243565Z"
  description: terraform backend key
  key_id: YCAJEftGSZSIPbfd2DWfPjuvL
  secret: YCN_vFn.....WvGWFNQ2w3
```

добавил ключи в `personal.auto.tfvars` (чтобы не забыть, файл в `.gitignore`, закоментил):

```
1 AWS_ACCESS_KEY_ID="YCAJE5UVmGrMEMPMbx7r9tkj2"
2 AWS_SECRET_ACCESS_KEY="YCN_vFn.....WvGWFNQ2w3"
```

сами ключи добавил в переменные окружения терминала:

```
1 export AWS_ACCESS_KEY_ID="YCAJE5UVmGrMEMPMbx7r9tkj2"
2 export AWS_SECRET_ACCESS_KEY="YCN_vFn.....WvGWFNQ2w3"
```

Настроил backend в `providers.tf`. Для этого переписал файл `providers.tf`

```
1 terraform {
2   required_providers {
3     yandex = {
4       source = "yandex-cloud/yandex"
5     }
6     template = {
7       source = "hashicorp/template"
8       version = "~> 2.2"
9     }
10
11   required_version = ">1.12.0"
```

```

12 }
13
14 backend "s3" {
15     bucket = "netology-course-bucket-kva"
16     key     = "terraform.tfstate"
17     region  = "ru-central1"
18
19     use_lockfile = true
20
21     endpoints = {
22         s3 = "https://storage.yandexcloud.net"
23     }
24
25     skip_region_validation      = true
26     skip_credentials_validation = true
27     skip_requesting_account_id  = true
28     skip_s3_checksum            = true
29 }
30 }
31
32 provider "yandex" {
33     cloud_id      = var.cloud_id
34     folder_id     = var.folder_id
35     zone          = var.default_zone
36     service_account_key_file = pathexpand("~/authorized_key.json")
37 }

```

Переместил файл state в s3 бакет и проверил его наличие (пришлось делать `terraform apply` т.к. при копировании ветки, в `hw-04` применял `terraform destroy` и `state` файл был уничтожен и не попал в новую папку `hw-05` при копировании):

```

1 terraform init -migrate-state
2 yc storage s3api list-objects --bucket netology-course-bucket-kva

```

```
vboxuser@KVA:~/terraform/hw-05$ terraform init -migrate-state
```

Initializing modules...

Initializing provider plugins found in the configuration...

- Reusing previous version of yandex-cloud/yandex from the dependency lock file
- Reusing previous version of hashicorp/template from the dependency lock file
- Using previously-installed yandex-cloud/yandex v0.204.0
- Using previously-installed hashicorp/template v2.2.0

Initializing the backend...

Initializing provider plugins found in the state...

- Reusing previous version of hashicorp/template
- Reusing previous version of yandex-cloud/yandex
- Using previously-installed yandex-cloud/yandex v0.204.0
- Using previously-installed hashicorp/template v2.2.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
vboxuser@KVA:~/terraform/hw-05$ yc storage s3api list-objects --bucket netology-course-bucket-kva
```

contents:

```
- key: terraform.tfstate
  last_modified: "2026-05-26T18:16:45.224Z"
  etag: '"7ff527d9f7071d5688fbe4372e3bdcae"'
  size: "17638"
  owner:
    id: ajeg4nq635d5tm88f1ka
    display_name: ajeg4nq635d5tm88f1ka
  storage_class: STANDARD
name: netology-course-bucket-kva
max_keys: "1000"
key_count: "1"
request_id: 2b5255e03001e9e4
```

Бакеты

Имя бакета

Все типы доступа 

Имя  	Занято  	Кол-во объектов  	Доступ 
netology-course-bucket-kva	4.84 КБ / 1 ГБ 	1 объект	С авторизацией ...

Запустил в обоих терминалах `terraform apply` и сработал `lock file`

```
vboxuser@KVA:~/terraform/hw-05$ terraform apply
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-a"]: Refreshing state... [id=e9bfcpsqnaavh3qc48vu]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhm1qm8st9uddcr2els3]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmhpm5rdb0q96vnspt]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

~ update in-place

Terraform will perform the following actions:

```
# module.marketing_vm.yandex_compute_instance.vm[0] will be updated in-place
~ resource "yandex_compute_instance" "vm" {
  id           = "fhm1qm8st9uddcr2els3"
  ~ labels     = {
    ~ "project" = "marketing" -> "marketing!"
    # (1 unchanged element hidden)
  }
  name         = "develop-marketing-0"
  # (15 unchanged attributes hidden)

  # (6 unchanged blocks hidden)
}
```

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value:

```
vboxuser@KVA:~/terraform/hw-05$ terraform apply
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

```
⦿ vboxuser@KVA:~/terraform/hw-05$ terraform apply
```

Error: Error acquiring the state lock

Error message: operation error S3: PutObject, https response error StatusCode: 412, RequestID: dbb8d2a81ef0baa2, HostID: , api error PreconditionFailed: At least one of the pre-conditions you specified did not hold

Lock Info:

ID:	d4c12d7a-8002-ebbc-cf07-3a151b5dddc0
Path:	netology-course-bucket-kva/terraform.tfstate
Operation:	OperationTypeApply
Who:	vboxuser@KVA
Version:	1.15.3
Created:	2026-05-27 11:22:16.795508037 +0000 UTC
Info:	

Terraform acquires a state lock to protect the state from being written by multiple users at the same time. Please resolve the issue above and try again. For most commands, you can disable locking with the "-lock=false" flag, but this is not recommended.

```
⦿ vboxuser@KVA:~/terraform/hw-05$
```

Принудительная разблокировка state файла из терминала где сработала блокировка через ключ `force-unlock`

```
1 terraform force-unlock d4c12d7a-8002-ebbc-cf07-3a151b5dddc0
```

```
vboxuser@KVA:~/terraform/hw-05$ terraform apply
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-a"]: Refreshing state... [id=e9bfcpsqnaavh3qc48vu]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmlqm8st9uddcr2els3]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmhpma5rdb0q96vnspt]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

~ update in-place

Terraform will perform the following actions:

```
# module.marketing_vm.yandex_compute_instance.vm[0] will be updated in-place
~ resource "yandex_compute_instance" "vm" {
  id              = "fhmlqm8st9uddcr2els3"
  ~ labels        = {
    ~ "project" = "marketing" -> "marketing!"
    # (1 unchanged element hidden)
  }
  name            = "develop-marketing-0"
  # (15 unchanged attributes hidden)

  # (6 unchanged blocks hidden)
}
```

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value:

```
vboxuser@KVA:~/terraform/hw-05$ terraform apply
pre-conditions you specified did not hold
Lock Info:
ID:          d4c12d7a-8002-ebbc-cf07-3a151b5dddc0
Path:        netology-course-bucket-kva/terraform.tfstate
Operation:   OperationTypeApply
Who:         vboxuser@KVA
Version:     1.15.3
Created:     2026-05-27 11:22:16.795508037 +0000 UTC
Info:
```

Terraform acquires a state lock to protect the state from being written by multiple users at the same time. Please resolve the issue above and try again. For most commands, you can disable locking with the "-lock=false" flag, but this is not recommended.

```
• vboxuser@KVA:~/terraform/hw-05$ terraform force-unlock d4c12d7a-8002-ebbc-cf07-3a151b5dddc0
Do you really want to force-unlock?
Terraform will remove the lock on the remote state.
This will allow local Terraform commands to modify this state, even though it may still be in use. Only 'yes' will be accepted to confirm.
```

Enter a value: yes

Terraform state has been successfully unlocked!

The state has been unlocked, and Terraform commands should now be able to obtain a new lock on the remote state.

```
• vboxuser@KVA:~/terraform/hw-05$
```

Ссылки на документацию и практики по теме домашнего задания:

- Terraform Backend S3
 - **Официальная документация backend S3** - настройка, параметры, `use_lockfile`, endpoints:
<https://developer.hashicorp.com/terraform/language/backend/s3>

- **State Locking** - как работает блокировка, `force-unlock` :
<https://developer.hashicorp.com/terraform/language/state/locking>
 - **State Storage and Locking** - общая концепция хранения state:
<https://developer.hashicorp.com/terraform/language/state/backends>
 - Yandex Cloud
 - **Yandex Object Storage как Terraform backend** - специфика настройки для YC:
<https://yandex.cloud/ru/docs/tutorials/infrastructure-management/terraform-state-storage>
 - **Создание статических ключей доступа** (AWS_ACCESS_KEY_ID / AWS_SECRET_ACCESS_KEY):
<https://yandex.cloud/ru/docs/iam/operations/sa/create-access-key>
 - **Yandex Object Storage S3-совместимый API**: <https://yandex.cloud/ru/docs/storage/s3/>
 - Дополнительно
 - **Native S3 locking без DynamoDB** (начиная с Terraform 1.10+) статья с объяснением механизма `PreconditionFailed` :
<https://rafaelmedeiros94.medium.com/goodbye-dynamodb-terraform-s3-backend-now-supports-native-locking-06f74037ad39>
 - **Terraform force-unlock**: <https://developer.hashicorp.com/terraform/cli/commands/force-unlock>
-

Задание 3

1. Сделайте в GitHub из ветки 'terraform-05' новую ветку 'terraform-hotfix'.
2. Проверьте код с помощью tflint и checkov, исправьте все предупреждения и ошибки в 'terraform-hotfix', сделайте коммит.
3. Откройте новый pull request 'terraform-hotfix' --> 'terraform-05'.
4. Вставьте в комментарий PR результат анализа tflint и checkov, план изменений инфраструктуры из вывода команды terraform plan.
5. Пришлите ссылку на PR для ревью. Вливать код в 'terraform-05' не нужно.

Разбор решения задания 3:

Создал из ветки `terraform-05` новую ветку `terraform-hotfix`.

```
• vboxuser@KVA:~/terraform/hw-05$ git checkout -b terraform-hotfix
Switched to a new branch 'terraform-hotfix'
• vboxuser@KVA:~/terraform/hw-05$ git push origin terraform-hotfix
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'terraform-hotfix' on GitHub by visiting:
remote:   https://github.com/ArturPirozhkov/terraform/pull/new/terraform-hotfix
remote:
To https://github.com/ArturPirozhkov/terraform.git
 * [new branch]      terraform-hotfix -> terraform-hotfix
```

Установил инструменты `tflint`

```
• vboxuser@KVA:~/terraform/hw-05$ curl -s https://raw.githubusercontent.com/terraform-linters/tflint/master/install_linux.sh | bash
arch=amd64
os=linux_amd64
```

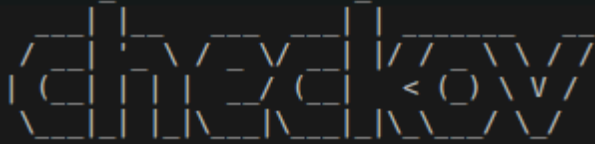
```
=====
Current tflint version
TFLint version 0.62.1
+ ruleset.terraform (0.14.1-bundled)
```

и `checkov`

```
• vboxuser@KVA:~/terraform/hw-05$ python3 -m venv .venv
• vboxuser@KVA:~/terraform/hw-05$ source .venv/bin/activate
• (.venv) vboxuser@KVA:~/terraform/hw-05$ pip install checkov
• (.venv) vboxuser@KVA:~/terraform/hw-05$ checkov -v
3.2.529
```

Запустил сканирование кода `checkov`

```
(.venv) vboxuser@KVA:~/terraform/hw-05$ checkov -d ~/terraform/hw-05 --compact
```



By Prisma Cloud | version: 3.2.529

terraform scan results:

Passed checks: 1, Failed checks: 6, Skipped checks: 0

Check: CKV_AWS_41: "Ensure no hard coded AWS access key and secret key exists in provider"

PASSED for resource: aws.default

File: /s3/providers.tf:8-15

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: marketing_vm

File: /main.tf:32-52

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: marketing_vm

File: /main.tf:32-52

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: analytics_vm

File: /main.tf:54-74

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: analytics_vm

File: /main.tf:54-74

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

FAILED for resource: s3

File: /s3/main.tf:1-7

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

FAILED for resource: s3

File: /s3/main.tf:1-7

secrets scan results:

Passed checks: 0, Failed checks: 2, Skipped checks: 0

Check: CKV_SECRET_6: "Base64 High Entropy String"

FAILED for resource: 24e7451df05ed5cd4cf1041be67c68f8d89d087a

File: /vault/terraform/main.tf:4-5

Check: CKV_SECRET_6: "Base64 High Entropy String"

- Terraform - 6 ошибок, все одного типа - модули подключены через ветку `?ref=main` вместо конкретного коммита или тега
- Secrets проверки - 2 ошибки, в `vault/terraform/main.tf` найдены строки похожие на Base64-секреты

Запустил сканирование кода `tflint` - `tflint --recursive`

```
vboxuser@KVA: ~/terraform/hw-05$ deactivate
vboxuser@KVA:~/terraform/hw-05$ tflint --recursive
11 issue(s) found:

Warning: Module source "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main" uses a default branch as ref (main) (terraform_module_pinned_source)

  on main.tf line 33:
 33:   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: Module source "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main" uses a default branch as ref (main) (terraform_module_pinned_source)

  on main.tf line 55:
 55:   source      = "git::https://github.com/udjin10/yandex_compute_instance.git?ref=main"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md

Warning: Missing version constraint for provider "yandex" in `required_providers` (terraform_required_providers)

  on providers.tf line 5:
 5:   yandex = {
 6:     source = "yandex-cloud/yandex"
 7:   }
   Follow link (ctrl + click)

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

Warning: Module source "github.com/terraform-yc-modules/terraform-yc-s3" is not pinned (terraform_module_pinned_source)

  on s3/main.tf line 2:
 2:   source = "github.com/terraform-yc-modules/terraform-yc-s3"

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_module_pinned_source.md
```

Warning: [Fixable] variable "default_cidr" is declared but not used (terraform_unused_declarations)

on variables.tf line 24:
24: variable "default_cidr" {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: [Fixable] variable "vpc_name" is declared but not used (terraform_unused_declarations)

on variables.tf line 30:
30: variable "vpc_name" {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: [Fixable] variable "vm_web_name" is declared but not used (terraform_unused_declarations)

on variables.tf line 46:
46: variable "vm_web_name" {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: [Fixable] variable "vm_db_name" is declared but not used (terraform_unused_declarations)

on variables.tf line 53:
53: variable "vm_db_name" {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_unused_declarations.md

Warning: terraform "required_version" attribute is required (terraform_required_version)

on vault/terraform/versions.tf line 1:
1: terraform {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_version.md

Warning: terraform "required_version" attribute is required (terraform_required_version)

```

on vpc/providers.tf line 1:
1: terraform {

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_version.md

Warning: Missing version constraint for provider "yandex" in `required_providers` (terraform_required_providers)

on vpc/providers.tf line 3:
3:     yandex = {
4:         source = "yandex-cloud/yandex"
5:     }

Reference: https://github.com/terraform-linters/tflint-ruleset-terraform/blob/v0.14.1/docs/rules/terraform_required_providers.md

vboxuser@KVA: ~/terraform/hw-05$

```

Сканер нашел 11 предупреждений (типа Warning):

1. Модули не закреплены по версии (3) - `main.tf` и `s3/main.tf` используют `?ref=main` вместо конкретного commit hash
2. Нет версии провайдера `yandex` (2) - в `providers.tf` и `vpc/providers.tf` не указана `version =`
3. Неиспользуемые переменные (4) - в `variables.tf` объявлены но нигде не используются
переменные: `default_cidr`, `vpc_name`, `vm_web_name`, `vm_db_name`
4. Не указана минимальная версия Terraform `required_version` ** (2) в `vault/terraform/versions.tf` и `vpc/providers.tf`

исправьте все предупреждения и ошибки в 'terraform-hotfix'

- получил хеши модулей:

```

1 git ls-remote https://github.com/udjin10/yandex_compute_instance.git HEAD
2 git ls-remote https://github.com/terraform-yc-modules/terraform-yc-s3.git HEAD

```

```

vboxuser@KVA:~/terraform/hw-05$ git ls-remote https://github.com/udjin10/yandex_compute_instance.git HEAD
f6638d3c9409a2ea7159c099c29052761e447 HEAD
vboxuser@KVA:~/terraform/hw-05$ git ls-remote https://github.com/terraform-yc-modules/terraform-yc-s3.git HEAD
64017d7de83fe105604fa7b612bc809a77c2fa2 HEAD

```

в файлах `main.tf` и `s3/main.tf` меняю источники удаленных модулей на полученные значения:

```

1 # main.tf
2
3 module "marketing_vm" {

```

```

4     source = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=f6638dd3c094d9a2ea7159cb096c29052761e447"
5     env_name      = "develop"
6     network_id    = module.vpc_dev.network_id
7     subnet_zones  = [var.default_zone]
8     subnet_ids    = [module.vpc_dev.subnets["ru-central1-a"].id]
9     instance_name  = "marketing"
10    instance_count = 1
11    image_family   = "ubuntu-2004-lts"
12    public_ip      = false
13
14    ---
15
16    module "analytics_vm" {
17        source = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=f6638dd3c094d9a2ea7159cb096c29052761e447"
18        env_name      = "develop"
19        network_id    = module.vpc_dev.network_id
20        subnet_zones  = [var.default_zone]
21        subnet_ids    = [module.vpc_dev.subnets["ru-central1-a"].id]
22        instance_name  = "analytics"
23        instance_count = 1
24        image_family   = "ubuntu-2004-lts"
25        public_ip      = false
26

```

```

1  # s3/main.tf
2
3  module "s3" {
4      source = "github.com/terraform-yc-modules/terraform-yc-s3?ref=e4017d77de83fe105604fa7b012bc809a77c2fa2"
5
6      bucket_name    = "netology-course-bucket-kva" # уникальное имя по всему облаку
7      force_destroy  = false                       # флаг не удалять с объектами

```

```
8     max_size      = 1073741824           # размер 1 ГБ в байтах
9 }
```

в файле `providers.tf` добавил версию: `version = ">= 0.204.0"`

```
1 yandex = {
2     source = "yandex-cloud/yandex"
3     version = ">= 0.204.0"
4 }
```

В файле `variables.tf` удалил (закомментировал) четыре неиспользуемые переменные:

- `variable "default_cidr"`
- `variable "vpc_name"`
- `variable "vm_web_name"`
- `variable "vm_db_name"`

В файле `vault/terraform/versions.tf` и `vpc/providers.tf` добавил `required_version`

```
1 terraform {
2     required_version = ">= 1.0.0"
3     required_providers {
4         vault = {
5             source = "hashicorp/vault"
6             version = "~> 3.0"
7         }
8     }
9 }
```

В файле `vpc/providers.tf` добавил версию `version = ">= 0.204.0"`:

```
1 terraform {
2   required_version = ">= 1.0.0"
3   required_providers {
4     yandex = {
5       source = "yandex-cloud/yandex"
6       version = ">= 0.204.0"
7     }
8   }
9 }
```

В файле `vault/main.tf` вынес креды в переменные

```
1 provider "vault" {
2   address          = "http://127.0.0.1:8200"
3   skip_tls_verify = true
4   token            = var.vault_token
5 }
6
7 data "vault_generic_secret" "vault_example" {
8   path = "secret/example"
9 }
10
11 resource "vault_generic_secret" "my_secret" {
12   path      = "secret/my_tf_secret"
13   data_json = jsonencode({
14     username = "admin"
15     password = var.vault_password
16   })
17 }
```

создал файл с переменными `vault/terraform/variables.tf`

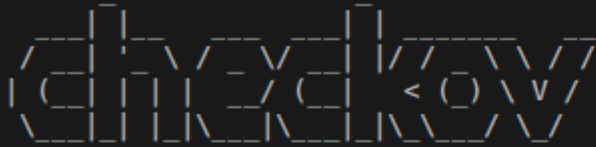
```
1 variable "vault_token" {
2     type          = string
3     description = "Vault root token"
4     sensitive     = true
5 }
6
7 variable "vault_password" {
8     type          = string
9     description = "Secret password for vault"
10    sensitive     = true
11 }
```

создал файл с переменнымси `vault/terraform/personal.auto.tfvars`

```
1 vault_token    = "education"
2 vault_password = "superpassword123"
```

Запустил сканеры:

```
(.venv) vboxuser@KVA:~/terraform/hw-05$ checkov -d ~/terraform/hw-05 --compact
```



By Prisma Cloud | version: 3.2.529

terraform scan results:

Passed checks: 7, Failed checks: 0, Skipped checks: 0

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

PASSED for resource: marketing_vm

File: /main.tf:32-52

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

PASSED for resource: marketing_vm

File: /main.tf:32-52

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

PASSED for resource: analytics_vm

File: /main.tf:54-74

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

PASSED for resource: analytics_vm

File: /main.tf:54-74

Check: CKV_TF_1: "Ensure Terraform module sources use a commit hash"

PASSED for resource: s3

File: /s3/main.tf:1-7

Check: CKV_TF_2: "Ensure Terraform module sources use a tag with a version number"

PASSED for resource: s3

File: /s3/main.tf:1-7

Check: CKV_AWS_41: "Ensure no hard coded AWS access key and secret key exists in provider"

PASSED for resource: aws.default

File: /s3/providers.tf:8-15

```
○ (.venv) vboxuser@KVA:~/terraform/hw-05$
```

```
● vboxuser@KVA:~/terraform/hw-05$ tflint --recursive
```

```
○ vboxuser@KVA:~/terraform/hw-05$
```

Ошибки ушли

Сделал `terraform plan`

```
• vboxuser@KVA:~/terraform/hw-05$ terraform plan
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.vpc_dev.yandex_vpc_network.this: Refreshing state... [id=enp2g2gun1aj9ca2fd02]
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd82mpu302omjlftrpsn]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd82mpu302omjlftrpsn]
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-a"]: Refreshing state... [id=e9bfcpsqnaavh3qc48vu]
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-b"]: Refreshing state... [id=e2l0r5shs28tjkfoq13r]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmhpm5rdb0q96vnspt]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmlqm8st9uddcr2els3]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
○ vboxuser@KVA:~/terraform/hw-05$
```

Результаты добавил в Pull Requests

<https://github.com/ArturP1rozhkov/terraform/pull/3>

Задание 4

1. Напишите переменные с валидацией и протестируйте их, заполнив default верными и неверными значениями. Предоставьте скриншоты проверок из terraform console.
 - type=string, description="ip-адрес" — проверка, что значение переменной содержит верный IP-адрес с помощью функций cidrhost() или regex(). Тесты: "192.168.0.1" и "1920.1680.0.1";
 - type=list(string), description="список ip-адресов" — проверка, что все адреса верны. Тесты: ["192.168.0.1", "1.1.1.1", "127.0.0.1"] и ["192.168.0.1", "1.1.1.1", "1270.0.0.1"].

Разбор решения задачи 4

Создал файл `variables_validation.tf`

```
1  variable "ip_address" {
2      type          = string
3      description = "ip-адрес"
4      default       = "192.168.0.1" # сначала верное значение затем неверное – "1920.1680.0.1"
5
6      validation {
7          condition      = can(cidrhost("${var.ip_address}/32", 0))
8          error_message = "Переменная ip_address должна содержать корректный IP-адрес. Например: 192.168.0.1"
9      }
10 }
11
12 variable "ip_address_list" {
13     type          = list(string)
14     description = "список ip-адресов"
15     default       = ["192.168.0.1", "1.1.1.1", "127.0.0.1"] # сначала верное значение, затем неверное –
16     ["192.168.0.1", "1.1.1.1", "1270.0.0.1"]
17
18     validation {
19         condition = alltrue([
20             for ip in var.ip_address_list :
21             can(cidrhost("${ip}/32", 0))
22         ])
23         error_message = "Все элементы списка ip_address_list должны быть корректными IP-адресами."
24     }
25 }
```

С верными значениями:

```
vboxuser@KVA:~/terraform/hw-05$ terraform init
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
vboxuser@KVA:~/terraform/hw-05$ terraform console
> var.ip_address
"192.168.0.1"
> var.ip_address_list
tolist([
  "192.168.0.1",
  "1.1.1.1",
  "127.0.0.1",
])
> ^C
```

С неверными:

```
vboxuser@KVA:~/terraform/hw-05$ terraform console
```

Error: Invalid value for variable

on variables_validation.tf line 1:

1: variable "ip_address" {

var.ip_address is "1920.1680.0.1"

Переменная ip_address должна содержать корректный IP-адрес. Например: 192.168.0.1

This was checked by the validation rule at variables_validation.tf:6,1-11.

Error: Invalid value for variable

on variables_validation.tf line 15:

15: variable "ip_address_list" {

var.ip_address_list is list of string with 3 elements

Все элементы списка ip_address_list должны быть корректными IP-адресами.

This was checked by the validation rule at variables_validation.tf:20,3-13.

Warning: Due to the problems above, some expressions may produce unexpected results.

> █

Переменная `ip_address`

```
1 variable "ip_address" {
2   type      = string      # тип - строка, например "192.168.0.1"
```

```
3 description = "ip-адрес"      # описание
4 default     = "192.168.0.1"  # значение по умолчанию
```

```
1 validation {
2   condition = can(cidrhost("${var.ip_address}/32", 0))
```

1. "\${var.ip_address}/32"

- Подставляет значение переменной в строку
- Из "192.168.0.1" делает "192.168.0.1/32"
- /32 - маска подсети, означает один конкретный хост (все 32 бита - адрес), хостов нет. Это стандартный способ записать "один конкретный IP" в CIDR-формате, значение 32 подставляю в виде интерполяции
- Нужна потому что `cidrhost()` принимает только CIDR-нотацию, а не чистый IP

2. `cidrhost("192.168.0.1/32", 0)`

- Встроенная функция Terraform для работы с сетями. Сеть из 1 адреса, только сам 192.168.0.1.
- Вычисляет IP-адрес хоста с номером 0 внутри сети 192.168.0.1/32
- Если IP **корректный**, то возвращает "192.168.0.1"
- Если IP **некорректный**, то **возвращает ошибку**

3. `condition` в блоке `validation` это обязательный аргумент, который принимает **булевое выражение** (true/false).

Правила condition:

- Должен возвращать **только** `true` или `false`
- Если `true` то **валидация прошла**, всё продолжается
- Если `false` то **Terraform останавливается** и выводит `error_message`
- Внутри можно использовать **только** `var.<имя_этой_переменной>` и нельзя ссылаться на другие переменные или ресурсы

4. `can(...)`

- Перехватывает ошибку от `cidrhost()`
- Если `cidrhost()` отработал без ошибки → `can()` возвращает `true`
- Если `cidrhost()` упал с ошибкой → `can()` возвращает `false`

```
1 error_message = "Переменная ip_address должна содержать корректный IP-адрес." } }
```

Это сообщение выводится когда `condition = false`.

Переменная `ip_address_list`

```
1 variable "ip_address_list" {
2   type = list(string) # тип - список строк
3   default = ["192.168.0.1", "1.1.1.1", "127.0.0.1"]
}
```

Блок validation:

```
1 validation {
2   condition = alltrue([
3     for ip in var.ip_address_list :
4     can(cidrhost("${ip}/32", 0))
5   ])
}
```

1. `for ip in var.ip_address_list`

- Цикл - проходит по каждому элементу списка
- На каждой итерации `ip` = очередной адрес, например `"1.1.1.1"`

2. `can(cidrhost("${ip}/32", 0))`

- Та же логика что и для одиночного IP
- Для каждого IP возвращает `true` или `false`

- В итоге цикл формирует список булевых значений: `[true, true, true]`

3. `alltrue([true, true, true])`

- Встроенная функция - возвращает `true` только если **все** элементы списка = `true`
- Если хотя бы один IP неверный, то список содержит `false`. Значит `alltrue()` = `false` и срабатывает `error_message`

Ссылки на документацию и практики:

- <https://developer.hashicorp.com/terraform/language/functions/cidrhost>
- <https://developer.hashicorp.com/terraform/language/functions/can>
- <https://developer.hashicorp.com/terraform/language/functions/regex>
- <https://developer.hashicorp.com/terraform/language/functions/alltrue>
- <https://developer.hashicorp.com/terraform/language/expressions/for>
- <https://dev.to/jajera/terraform-validation-rules-best-practices-examples-4abi>
- <https://stackoverflow.com/questions/71143820/how-do-you-validate-ip-addresses-in-terraform/76072899>
- <https://georgedavis-triumphtech.github.io/terraform/2023/01/20/terraform-input-validation-samples/>

Дополнительные задания (со звёздочкой*)

Задание 5*

1. Напишите переменные с валидацией:

- `type=string, description="любая строка"` — проверка, что строка не содержит символов верхнего регистра;
- `type=object` — проверка, что одно из значений равно `true`, а второе `false`, т. е. не допускается `false false` и `true true`:

```
1 variable "in_the_end_there_can_be_only_one" {
2     description="Who is better Connor or Duncan?"
3     type = object({
4         Dunkan = optional(bool)
```

```

5      Connor = optional(bool)
6  })
7
8  default = {
9      Dunkan = true
10     Connor = false
11 }
12
13 validation {
14     error_message = "There can be only one MacLeod"
15     condition = <проверка>
16 }
17 }

```

Разбор решения задания 5*

По аналогии с предыдущим, добавил в файл `variables_validation.tf`

```

1  variable "no_uppercase_string" {
2      type      = string
3      description = "любая строка"
4      default    = "hello world"
5
6      validation {
7          condition      = var.no_uppercase_string == lower(var.no_uppercase_string)
8          error_message = "Строка не должна содержать символы верхнего регистра."
9      }
10 }

```

- `lower(var.no_uppercase_string)` - встроенная функция, переводит строку в нижний регистр
- Сравнивает оригинал с нижним регистром через знак двойное =
- Если строка уже в нижнем регистре, то они равны, значит `true`

- Если есть заглавные, они не равны и значит `false`

```
soxuser@KVA:~/terraform/hw-05$ terraform console
on variables validation.tf line 30:
30: variable "no_uppercase_string" {
    var.no_uppercase_string is "Hello world"

Строка не должна содержать символы верхнего регистра.
This was checked by the validation rule at variables_validation.tf:35,3-13.
```

Далее следующий блок туда же:

```
1 variable "in_the_end_there_can_be_only_one" {
2   description = "Who is better Connor or Duncan?"
3   type = object({
4     Dunkan = optional(bool)
5     Connor = optional(bool)
6   })
7
8   default = {
9     Dunkan = true
10    Connor = false
11  }
12
13  validation {
14    error_message = "There can be only one MacLeod"
15    condition = (
16      var.in_the_end_there_can_be_only_one.Dunkan != var.in_the_end_there_can_be_only_one.Connor
17    )
18  }
19 }
```

Здесь:

- `!=` - оператор "не равно"
- Если значения **разные** (один `true`, другой `false`), то `true`
- Если значения **одинаковые** (оба `true` или оба `false`), то `false`

Запускаю:

```
1 # Верное значение – пройдёт
2 terraform plan -var='in_the_end_there_can_be_only_one={"Dunkan":true,"Connor":false}'
3
4 # Неверное – оба true, должна быть ошибка
5 terraform plan -var='in_the_end_there_can_be_only_one={"Dunkan":true,"Connor":true}'
6
```

```
vboxuser@KVA:~/terraform/hw-05$ terraform plan -var='in_the_end_there_can_be_only_one={"Duncan":true,"Connor":true}'
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.vpc_dev.yandex_vpc_network.this: Refreshing state... [id=enp2g2gun1aj9ca2fd02]
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd82mpu302omjlftrpsn]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd82mpu302omjlftrpsn]
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-b"]: Refreshing state... [id=e2l0r5shs28tjkfoq13r]
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-a"]: Refreshing state... [id=e9bfcpsqnaavh3qc48vu]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmhpmma5rdb0q96vnspt]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmlqm8st9uddcr2els3]
```

Planning failed. Terraform encountered an error while generating this plan.

Error: Invalid value for variable

```
on variables_validation.tf line 42:
42: variable "in_the_end_there_can_be_only_one" {
    |
    |   var.in_the_end_there_can_be_only_one.Connor is true
    |   var.in_the_end_there_can_be_only_one.Duncan is true
```

There can be only one MacLeod

This was checked by the validation rule at variables_validation.tf:54,3-13.

```
vboxuser@KVA:~/terraform/hw-05$ █
```